Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	States the correct binomial distribution	AO3.3	B1	B(6, 0.15)
(b)	Calculates the correct probability	AO1.1b	B1	0.0000114
(c)	Calculates P(X ≤ 1) or P(X ≤ 2) using the binomial distribution	AO1.1a	M1	P(X ≤ 1) = 0.7764 P(X ≥ 2) = 1 − P(X ≤ 1) P(X ≥ 2) = 0.224
	Obtains the correct answer	AO1.1b	A1	
(d)	Finds the correct mean	AO1.1b	B1	0.9
(e)	States a first appropriate assumption in context	AO3.5b	B1	The probability of a light bulb being faulty is fixed
	States a second appropriate assumption in context	AO3.5b	B1	A light bulb being faulty is independent of any other light bulb being faulty
				Total 7 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)(i)	Obtains the correct mean	AO1.1b	B1	1.38
(ii)	Uses the correct formula for standard deviation	AO1.1a	M1	$\sqrt{\frac{261.8}{120}-1.38^2}$ 0.526 to 0.529
	Obtains the correct standard deviation	AO1.1b	A1	
(b)(i)	Uses the model to calculate a normal probability	AO3.4	M1	0.5417 to 0.5428
	Obtains correct probability	AO1.1b	A1	
(ii)	Recalls correct value of 0	AO1.2	B1	0
(c)	Calculates the value of mean − 3 × standard deviations	AO1.1b	M1	−0.1998 to −0.207

Concludes that model might be inappropriate as the value is less than 0	AO2.2b	A1	This is less than 0 and so model might not be appropriate
(d) Standardises appropriately and formalises a probability statement PI by fully correct equation	AO3.1b	M1	$P\left(Z > \frac{0.75-\mu}{0.21}\right) = 0.1$ z value = 1.2816 $\frac{0.75-\mu}{0.21} = 1.2816$ μ = 0.481
Obtains z value from inverse normal distribution (±1.2816)	AO1.1a	M1	
Forms a correct equation using standardised result and z value	AO1.1b	A1	
Solves the equation to find the correct value of μ	AO1.1b	A1	
			Total 12 marks

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds value for p	AO1.1a	M1	$p = \frac{1680}{2400} = 0.7$
	Finds correct probability from calculator	AO1.1b	A1	Using $X \sim$ B (25,0.7), P(X = 22) = 0.0243
(b)	Explains the reason why the model may no longer apply in context	AO3.5b	E1	It is likely that all the houses (and gardens) will be of similar types, and hence similar owners, so not likely to be independent as binomial model requires.
				Total 3 marks

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	States both hypotheses correctly for one-tailed test	AO2.5	B1	X = number of Christmas holidays without illness since January 2007

			$X \sim B(7, p)$ $H_0\ p = 0.65$ $H_1\ p < 0.65$
States model used (condone 0.009 rather than 0.056) **PI**	AO1.1b	M1	Under null hypothesis, $X \sim B(7, 0.65)$
Using calculator, 0.056 or better	AO1.1b	A1	$P(X \leq 2) = 0.0556$
Evaluates binomial model by comparing $P(X \leq 2)$ with 0.05 **PI**	AO3.5a	M1	$0.0556 > 0.05$
Infers H_0 accepted PI	AO2.2b	A1	Accept H_0
Concludes correctly in context. 'not sufficient evidence' or equivalent required	AO3.2a	E1	There is not sufficient evidence that John's rate of illness has decreased

(b)

States one correct assumption(s) regarding validity of model	AO3.5b	E1	**Assumption 1** The probability of illness remains constant throughout one's life **Validity** Not fully valid, as age has an impact on the immune system **OR** **Assumption 2** Annual results (of illness) are independent of one another **Validity** (Largely) valid. Trials are sufficiently far apart that an illness spanning two Christmases is unlikely. **OR** **Assumption 3** There are only two states, well and ill **Validity** Unclear. Grey area exists. eg does a mild sore throat count as ill?
States corresponding correct description(s) of likelihood of validity in context	AO2.4	E1	
States second correct assumption(s) regarding validity of model	AO3.5b	E1	
States corresponding correct description(s) of likelihood of validity in context	AO2.4	E1	
Max two assumptions with description of validity			
			Total 10 marks

Q5.

(a) $\sigma \approx \frac{10}{a}$ or $\frac{20}{b}$ or $\frac{\text{range}}{b}$ **or** 10c **or** 20d

OE; with $2 \le a \le 4$ $\quad 4 \le b \le 8$
or with c *or* d *in equiv percentages*
***Cannot** be implied from a correct answer (justification required)*

M1

2.5 or 3.3(OE) or 5

A1

2

SC Award B1 for only **2.5 or 3.3(OE)** or 5 with no justification
Award B0 for any other answer with no justification or with incorrect justification
(eg √10 = 3.16)

(b) **Valid statement** involving:
391 and 405
OR
401 and 415
OR
24 and 10
OR
391 and 415 and 10 / 24 with linking statement

Allow 'set weight' to imply 415
and / or 'mean' to imply 391
B0 for 10 linked to σ

B1

95.5 > (**value** of σ of **2.5 or 3.3(OE) or 5**)

Accept ≠ rather than >
***Clear correct numerical** comparison*

B1

Neither (likely to be) **correct**

Dependent on B1 B1

Bdep1

3

(c) Mean **or** $\bar{y} = \frac{8210.0}{10}$ = **821**

CAO;

OR

$\sum y$ = **<u>8200</u>**

B1

Variance $\frac{110.00}{9}$ = **<u>12.2</u>**

AWRT

or

$\frac{110.00}{10}$ = **<u>11</u>**

CAO
*Award on **value**; ignore notation*

OR

SD **<u>3.5 or 3.3</u>**

AWRT

B1

821 **is similar to / within 10 of** 820

*OE; **clear correct numerical** comparison of **821** with **820***
Allow 'set weight' to imply 820

OR

Or

8210 **is within 100 of** 8200

*OE; **clear correct numerical** comparison of **8210** with **8200** but do **not** accept 'within 10' here*

B1

3.5 **or** 3.3 is
similar to
a value of σ of **3.3(OE) or 2.5**

***Clear correct numerical** comparison*

B1 4

[9]

Q6.

(a) (i) Volume, $X \sim \mathrm{N}(32, 10^2)$

$P(X < 40) = P\left(Z < \frac{40 - 32}{10}\right)$

Standardising 40 with 32 and 10; allow (32 − 40)

M1

$= P(Z < 0.8)$

CAO; ignore inequality and sign
May be implied by a correct answer

A1

= 0.788

AWRT (0.78814)

A1

3

(ii) $P(X > 25) = P(Z < -0.7) = P(Z < +0.7)$

Area change
May be implied by a correct answer or an answer > 0.5

M1

= 0.758

AWRT (0.75804)

A1

2

(iii) $P(25 < X < 40) =$ (i) − (1 − (ii)

OE; allow new start ignoring (i) & (ii)
Allow even if incorrect standardising providing 0 < answer < 1
*May be implied by a **correct** answer*

M1

= 0.78814 − (1 − 0.75804) = 0.546

Note:

If (ii) is 0.242, then (0.788 − 0 242) = 0.546 ⇒ M0 A0

AWRT (0.54618)

A1

2

(b) $P(B < £65) = P\left(Z > \frac{48.5 - 32}{10}\right)$

Attempt to change from B to X using (48 to 49), 32 and 10 or P$\left(Z > \frac{65 - 42.88}{13.4}\right)$

Attempt to work with distribution of B
using 65, (42.8 to 42.9) and 13.4

M1

= P(Z < 1.65) = 1 − P(Z < 1.65)

Area change
May be implied by a correct answer or an answer < 0.5

m1

= 1 − 0.95053 = 0.049 to 0.05(0)

AWFW (0.04947)

A1

3

(c) Other **fuels**
Other **vehicles** with an example (not other cars)
Other types of **customer**
Minimum purchase (policy)
Purchases in integer / fixed £s
Customers filling fuel cans

Size of car / engine / fuel tank ⇒ B0
Price of fuel ⇒ B0
Customer paying capacity ⇒ B0
*Must be **two clearly different** valid reasons for award of B2*
Drivers and vehicles related ⇒ B1 eg lorry drivers & lorries

B2,1

2

[12]

Q7.

(a) $\underline{U \sim \mathrm{B}(40, 0.15)}$

Used somewhere in (a)

M1

(i) P(U = 6) = 0.6067 − 0.4325
or

= $\binom{40}{6}$ $(0.15)^6$ $(0\ 0.85)^{34}$

Accept 3 dp rounding or truncation

M1

= 0.174

Can be implied by a correct answer

AWRT *(0.1742)*

A1

3

(ii) P(U ≤ 5) = 0.432 to 0.433

AWFW *(0.4325)*

B1

1

(iii) P(5 < U < 10) = 0.9328 or 0.9701 (p1)

Accept 3 dp rounding or truncation but allow 0.97

M1

$p_2 - p_1$ ⇒ *M0 M0 A0*

(1 − p_2*)* − p_1 ⇒ *M0 M0 A0*

p_1 *− (1 −* p_2*)* ⇒ *M0 M0 A0*

(1 − p_2*) − (1 −* p_1*)* ⇒ *M1 M1 (A1)*

only providing result > 0

MINUS 0.4325 or 0.2633 (p2)

Accept 3 dp rounding or truncation

M1

= 0.5(00) to 0.501

AWFW *(0.5003)*

A1

3

Alternative solution

B(40, 0.15) expressions stated for at least 3 terms within 5 ≤ U ≤ 10 gives probability

Can be implied by a correct answer

(M2)

= 0.5(00) to 0.501

AWFW *(0.5003)*

(A1)

u	(5)	6	7	8	9	(10)
P(U = u)	(0.1692)	0.1742	0.1492	0.1087	0.0682	(0.0373)

(3)

(b) Mean or μ = 32 × 0.15 = 4.8

CAO

B1

(V or σ^2 =) <u>32 × 0.15× 0.85</u>
or
(SD or σ =) $\underline{\sqrt{32 \times 0.15 \times 0.85}}$

Either numerical expression; ignore terminology
May be implied by 4.08 CAO seen or 2.02 AWRT seen

M1

(SD or σ) = 2.02

AWRT *(2.0199)*
Do not award if labelled V or σ^2

A1

3

(c) Mean = 7.7

CAO *($\sum x$ = 77)*

B1

SD = 1.26 to 1.34

AWFW *($\sum x^2$ = 609)*

B1

(Sample) mean is bigger / greater / different or 7.7 / 32 = 0.24 > 0.15
and
(Sample) SD is smaller / less / different

Both; dependent on all previous 5 marks of B1 M1 A1 B1 B1
Can be scored for incorrect (b) re-done correctly in (c)
Means & SDs different → Bdep0

Bdep1

So model appears unsuitable

OE; dependent on Bdep1

Bdep1

4

[14]

Q8.

(a) (i) $\underline{U \sim \text{B}(30,\ 0.13,\ 0.35 \text{ or } 0.20)}$

Used correctly anywhere in (a)

M1

$\text{P}(P = 2) = \binom{30}{2}(0.13)^2\,(0.87)^{28}$

*Can be implied by a **correct** answer*

A1

= **<u>0.148 to 0.15</u>**

AWFW *(0.1489)*

A1

3

(ii) p = **<u>0.35</u>**

CAO

B1

$\text{P}(R \cup P > 10)$ = **<u>1 – (0.5078 or 0.3575)</u>**

Requires '1 –'
Accept 3 dp rounding or truncation
Can be implied by 0.49 to 0.493
*but **not** by 0.642 to 0.643*

M1

= **<u>0.49 to 0.493</u>**

AWFW *(0.4922)*

A1

3

(iii) $\text{P}(5 \le G \le 10)$ = **0.9744 or 0.9389**(p_1)

Accept 3 dp rounding or truncation

M1

MINUS 0.2552 or 0.4275 (p_2)

Accept 3 dp rounding or truncation

M1

= **<u>0.719 to 0.72</u>** (p_3)

AWFW *(0.7192)*

A1

3

Notes **1** $p_3 \le 0$ or $p_3 \ge 1 \Rightarrow$ M0 M0 A0

2 $p_2 - p_1 \;\Rightarrow$ M0 M0 A0
3 $(1 - p_2) - p1 \;\Rightarrow$ M0 M0 A0
4 $p_1 - (1 - p_2) \Rightarrow$ M1 M0 A0
5 $p_1 \times p_2 \;\Rightarrow$ M1 M0 A0
6 $(1 - p_2) - (1 - p_1) \Rightarrow$ M1 M1 (A1)

(b) (i) Mean or μ = 100 × 0.22 = **22**
Variance or σ^2 = 100 × 0.22 × 0.78

CAO

B1

= **17.1 to 17.2**

AWFW (ignore notation) (17.16)
ISW all subsequent working

B1

2

(ii) 22.1 ≈ / = 22 or means similar / equal **or**
0.221 ≈ / = 0.22 or proportions similar / equal so
reject claim (that p > 0.22)
or
accept that p = 0.22

Dependent on 22 seen in (b)(i) or (ii)
Accept diff = 0.1 CAO
***Correct** (numerical)*
*comparison with **correct** conclusion (even if at end and stated as 'reject (both) claims')*

B1

$\sqrt{17.1 \text{ to } 17.2}$ = **4.13 to 4.15 ≈ / = 4.17**

*Comparison using two values **or***
one value + diff (0.02 to 0.04 AWFW)

or

17.1 to 17.2 ≈ / = 17.3 to 17.4

*Comparison using two values **or***
one value + diff (0.1 to 0.3 AWFW)

B1

so
reject claim that not random samples
or
accept that are random samples

Dependent on previous B1
***Correct** conclusion regarding randomness of sample*

Bdep1

3

[14]

Q9.

(a) (i) R ~ B (14, 0.35)

Used somewhere in (a); may be implied

M1

P(R ≤ 7) = 0.924 to 0.925

AWFW (0.92466)

A1

2

(ii) P(R ≥ 11) = 1 – P(R ≤ 10)

= 1 – (0.9989 or 0.9999)

Requires '1 –' and ≥ 4 dp accuracy

M1

= 0.0011

AWRT (0.001106)

A1

2

(iii) P(5 < R < 10) = 0.9940 or 0.9989 (p_1)

Accept 3 dp accuracy

$p_2 - p_1 \Rightarrow$ *M0 M0 A0*

$(1 - p_2) - p_1 \Rightarrow$ *M0 M0 A0*

$p_1 - (1 - p_2) \Rightarrow$ *M1 M0 A0*

only providing result > 0

M1

minus 0.6405 or 0.4227 (p_2)

Accept 3 dp accuracy

M1

= 0.353 to 0.354

AWFW (0.35346)

A1

or

B(14, 0.35) expressions stated for **at least 3** terms within 4 ≤ R ≤ 11 gives probability

Can be implied by correct answer

(M1)

= 0.353 to 0.354

AWFW (0.35346)

(A2)

3

(b) $R \sim B(21, 0.35)$

Implied from correct stated formula; do not accept misreads

M1

$P(R = 4) = \binom{21}{4}(0.35)^4(0.65)^{17}$

Can be implied by a correct answer
Ignore any additional terms

A1

= 0.059 to 0.0595

AWFW (0.059274)

A1

3

(c) (i) $S \sim B(7, 5/7)$

Mean = np = 7 × 5/7 = 5

If not identified, assume order is μ then σ^2

CAO

B1

Variance = $np(1 - p)$
= 7 × 5/7 × 2/7 = 10/7 or 1.42 to 1.43

Must clearly state variance value if standard deviation (also) stated
CAO / AWFW

B1

2

(ii) **Means** are the **same**
and (both comparisons clearly stated)
Variances/standard deviations are **similar**
Do not accept statements involving correct/incorrect/exact/etc

Must have scored B1 B1 in (i) or B1 B0 plus

10/7 v 1.5 or $\sqrt{10/7}$ v $\sqrt{1.5}$ ***stated***

B1dep

Barry's claim appears/is
sound/valid/correct/likely

Must have scored previous B1dep

B1dep

2

[14]

Q10.

(a) (i) B(16 or 25 or 40, 0.45)

Used at least once in (a)(i) to (iii)

M1

$$P(S = 3) = \binom{16}{3}(0.45)^3(0.55)^{13}$$

May be implied by correct answer
Ignore any additional terms

A1

= 0.021 to 0.022

AWFW (0.0215)

A1

3

(ii) P(S < 10) = 0.3843 or 0.2424

Accept 3 dp accuracy from tables or calculation

B1

= 0.242 to 0.243

AWFW (0.2424)

B1

2

(iii) P(15 ≤ S ≤ 20) = 0.7870 or 0.6844 (p_1)

Accept 3 dp accuracy

M1

$p_2 - p_1 \Rightarrow$ *M0 M0 A0*
$p_1 - (1 - p_2) \Rightarrow$ *M1 M0 A0*

minus 0.1326 or 0.2142 (p_2)

Accept 3 dp accuracy / truncation

M1

= 0.654 to 0.655

AWFW (0.6544)

A1

OR

B(40, 0.45) expressions stated for at least 3 terms within $14 \le S \le 20$ gives probability

Or implied by a correct answer

(M1)

= 0.654 to 0.655

AWFW

(A2)

3

(iv) Mean, $\mu = np = 50 \times 0.45 = 22.5$ or 22½

CAO (22.5 = 22 or 23 $\Rightarrow$ ISW)

B1

Variance, $\sigma^2 = np(1 - p)$

Accept $12\,\frac{3}{8}$ *or* $\frac{99}{8}$

= 50 × 0.45 × 0.55

= 12.3 to 12.4

AWFW (12.375)

B1

2

(b) (i) **Non-independence** of senior citizens travel

Senior citizens tend to **travel in pairs/groups**

Or equivalent; but must be a clear indication of non-independent events

B1

1

(ii) 7.15 am is outside 9.30 am to 11.30 am

Cannot use SCPs before 9.30 am

Cannot use SCPs @ 7.15 am

Cannot use SCPs during morning 'rush hour'

Value of p likely to be smaller/different/zero

Data not available

Or equivalent

*Accept other **sensible** reasons*

B1

Senior citizens not out at this time
Passengers likely to be workers/school children

*Distribution of **types of** passenger different*

1

[12]

Q11.

(a) Mean = $\frac{\sum fx}{\sum f} = \frac{275}{99} = 2.77 \text{ to } 2.78$

AWFW (2.778)

B1

If not identified, assume order is $\bar{x}$ then *s*

Treat rounding to integers as ISW

SD $\left(\sum fx^2 = 933\right) = 1.3(0) \text{ to } 1.32$

AWFW (1.307 & 1.314)

B2

Special Case:

Evidence of $\frac{\sum fx}{99}$

Can award if no marks scored in (ii)

(M1)

3

(b) (i) Mean$_{163}$ = $\frac{99 \times \text{Mean}_{99}}{163}$ or $\frac{\sum fx \text{ from(a)(ii)}}{163}$

Or equivalent; may be implied by an answer within range

M1

= 1.68 to 1.69

AWFW (1.687)

A1

2

(ii) Increase

CAO; or equivalent (1.696)

Ignore any working (1.702)

B1

1

(iii) Data is (positively/negatively) skewed / not symmetric / bimodal / not bell-shaped from frequency distribution / given table

Or equivalent

B1

or

[C's mean in (b)(i)] – 2 × [C's SD in (a) (ii)] <

(–1.75 to –0.90)

or

[C's mean in (b)(i)] – 2 × [1.69 to 1.71] < 0

Thus claim appears **not valid**

Or equivalent

Dependent upon previous B1

B1dep

2

[8]

Q12.

(a) (i) Median (50) = **3**

CAO

Do not award marks if correct answers are based on shown incorrect method; eg accept use of 99/2, etc but not 276/2, etc

B1

If not identified, then assume order is median then IQR

IQR (75 – 25) = 4 – 2 = **2**

CAO; but

25th value $\Rightarrow$ *IQR = 2* $\Rightarrow$ *B0*

B2

Special Cases:

Identification that LQ = 2 and UQ = 4

Both CAO

(B1)

Statement of ≥ 4 cumulative frequencies

F: 14 49 74 87 96 98 99

Can award if no marks scored in (i)
even if then applied to continuous data

(M1)

3

(ii) Mean = $\frac{\sum fx}{\sum f} = \frac{275}{99} = 2.77 \text{ to } 2.78$

AWFW (2.778)

B1

If not identified, assume order is $\bar{x}$ then *s*

Treat rounding to integers as ISW

SD $\left(\sum fx^2 = 933\right) = 1.3(0) \text{ to } 1.32$

AWFW (1.307 & 1.314)

B2

Special Case:

Evidence of $\frac{\sum fx}{99}$

Can award if no marks scored in (ii)

(M1)

3

(b) (i) $\text{Mean}_{163} = \frac{99 \times \text{Mean}_{99}}{163}$ or $\frac{\sum fx \text{ from (a)(ii)}}{163}$

Or equivalent; may be implied by
an answer within range

M1

= 1.68 to 1.69

AWFW (1.687)

A1

2

(ii) Increase

CAO; or equivalent (1.696)
Ignore any working (1.702)

B1

1

(iii) Data is (positively/negatively) skewed /
not symmetric / bimodal / not bell-shaped from

frequency distribution / given table

Or equivalent

B1

or

[C's mean in (b)(i)] – 2 × [C's SD in (a) (ii)] < 0

(–1.75 to –0.90)

or

[C's mean in (b)(i)] – 2 × [1.69 to 1.71] < 0

Thus claim appears **not valid**

Or equivalent

Dependent upon previous B1

B1dep

2

[11]

Q13.

(a) R ~ B(50, 0.15)

(i) P(R < 10) **= 0.791**

AWRT (0.7911)

B1

(ii) P(5 ≤ R ≤ 10) = 0.8801 or 0.7911 (p_1)

Accept 3 dp accuracy

$(1 – p_2) – p_1 \Rightarrow$ M0 M0 A0

$p_1 – (1 – p_2) \Rightarrow$ M1 M0 A0

only providing result > 0

M1

minus 0.1121 or 0.2194 (p_2)

Accept 3 dp accuracy

M1

= 0.768

AWRT (0.7680)

A1

or

B(50, 0.15) expressions stated for **at least 3** terms within 4 ≤ R ≤ 10 gives probability

Can be implied by correct answer

(M1)

= 0.768

AWRT

(A2)

4

(b) **Confusion of** 22, 35, 120 and/or 0.15, 0.06

*Do **not** treat as misreads*

(i) $S \sim \mathrm{B}(22, 0.06)$

Used in (b)(i) as evidenced by any correct binominal term for $S > 0$

M1

$$P(S = 2) = \binom{22}{2}(0.06)^2(0.94)^{20}$$

Can be implied by correct answer
Ignore any additional terms

A1

= 0.24 to 0.242

AWFW (0.24125)

A1

3

(ii) $P(S \geq 1) = 1 - q^{35}$ where $0.84 \leq q \leq 0.96$

Can be implied by correct answer
Award for $(0.94)^{35}$ seen in an expression but not if accompanied by a multiplier ≠ 1

M1(B1)

= 0.885 to 0.89

AWFW (0.88532)

A1

2

(iii) Mean = np = 120 × 0.94 = 112.8 or 113

Either

B1

If not identified, assume order is μ then σ^2

Variance = $np(1 - p)$

Must clearly state variance value

= 120 × 0.94 × 0.06 **=** 6.76 to 6.78

AWFW (6.768)

B1

2

(iv) **Means** are (approximately) the **same** stated

Must have scored 1st B1 in (iii)

or

Variances are (very) **different** stated

Must have scored 2nd B1 in (iii)

B1

Agree with P(sorts letter incorrectly) = 0.06

Dependent on 'means same' stated

B1dep

Disagree with independent from letter to letter

Dependent on 'variances different' stated

B1dep

3

[14]

Q14.

Binomial distribution

Used somewhere in question

M1

(a) $M \sim B(40, 0.35)$

Used; may be implied

A1

$P(M \leq 15) = 0.69(0)$ to 0.696

AWFW (0.6946)

A1

3

(b) $W \sim B(10, 0.29)$

Used; may be implied

B1

$$P(W = 3) = \binom{10}{3}(0.29)^3(0.71)^7$$

Stated; may be implied

M1

= 0.266 to 0.2665

AWFW (0.2662)

***Note**: B(10, 0.3)* $\Rightarrow$ *0.2668*

A1

3

(c) (i) $n = 20$ $p = 0.71$

Stated or used; may be implied by 14.2

B1

Mean, $\mu = np = 14.2$

CAO

B1

Variance, $\sigma^2 = np(1 - p)$

$= 4.11$ to 4.12

AWFW (4.118)

B1

3

(ii) Mean of 16.5 is greater/different **or**
16.5/20 = 0.825 is greater/different to 0.71

Dependent on μ = 14.2

B1dep

Means and variances are different

(B2,1 dep)

Variance of 2.50 is smaller/different

Dependent on σ^2 = 4.11 to 4.12

B1dep

Suggests **claim** that groups are not
random samples **is justified**

Dependent on previous 2 marks
Or equivalent

B1dep

3

[12]

Q15.

Binomial distribution

Used somewhere in question

M1

(a) (i) $M \sim B(40, 0.35)$

Used; may be implied

A1

$P(M \leq 15) = 0.69(0)$ to 0.696

AWFW (0.6946)

A1

3

(ii) $P(10 < M < 20) = 0.9637$ or 0.9827

Accept 3 dp accuracy

M1

minus 0.1215 or 0.0644

Accept 3 dp accuracy

M1

= 0.84(0) to 0.843

AWFW (0.8422)

A1

OR

B(40, 0.35) expressions stated for
at least 3 terms within $10 \leq M \leq 20$
Answer = 0.84(0) to 0.843

Or implied by a correct answer

(M1)

AWFW

(A2)

3

(b) $W \sim B(10, 0.29)$

Used; may be implied

B1

$P(W = 3) = \binom{10}{3}(0.29)^3(0.71)^7$

Stated; may be implied

M1

= 0.266 to 0.2665

AWFW (0.2662)

***Note**: B(10, 0.3) $\Rightarrow$ 0.2668*

A1

3

(c) (i) n = 20 p = 0.71

Stated or used; may be implied by 14.2

B1

Mean, $\mu = np$ = 14.2

CAO

B1

Variance, $\sigma^2 = np(1 - p)$ = 4.11 to 4.12

AWFW (4.118)

B1

3

(ii) Mean of 16.5 is greater/different **or** 16.5/20 = 0.825 is greater/different to 0.71

Dependent on μ = 14.2

B1dep

Means and variances are different

(B2,1 dep)

Variance of 2.50 is smaller/different

Dependent on σ^2 = 4.11 to 4.12

B1dep

Suggests **claim** that groups are not random samples **is justified**

Dependent on previous 2 marks
Or equivalent

B1dep

3

[15]

Q16.

(a) Use of binomial in (a), (b) or (c)

Can be implied

M1

$$P(E = 5) = \binom{16}{5}(p)^5(1-p)^{11}$$

Allow p = 0.45, 0.25, 0.30 or $\frac{1}{3}$

M1

= 0.112

AWRT (0.1123)

A1

3

(b) (i) B(50, 0.25)

Used; can be implied

B1

P(C ≤ 12) = 0.511

AWRT (0.5110)

B1

2

(ii) P(10 < B' < 20) = 0.9152 or 0.9522

Allow 3 dp accuracy

M1

minus 0.0789 or 0.1390

Allow 3 dp accuracy

M1

= 0.836

AWRT (0.8363)

A1

or

B (50, 0.30) expressions stated for **at least 3** terms within 10 ≤ B' ≤ 20

Or implied by a correct answer

(M1)

Answer = 0.836
AWRT

(A2)
3

(c) $n = 40$, $p = 0.7$
Both used; can be implied

B1

Mean $\mu = np = 28$
CAO; ft on p only

B1ft

Variance $\sigma^2 = np(1 - p) = 8.4$
Use of $np(1 - p)$ even if SD

M1

Standard deviation = $\sqrt{8.4}$
or = 2.89 to 2.9
CAO; AWFW

A1
4

[12]

Q17.

(a) Use of binomial in (a) or (b)(i)
PI

M1

(i) $P(T_{10} \leq 3) = 0.38$ to 0.383
AWFW (0.3823)

B1
2

(iii) $P(10 < T_{40} < 20) = 0.8702$ or 0.9256
Allow 3 dp accuracy

M1

minus 0.0352 or 0.0156

Allow 3 dp accuracy

M1

= 0.83 to 0.84

AWFW (0.835)

A1

OR

B (40, 0.40) expressions stated for at least 3 terms within 10 ≤ T_{40} ≤ 20

Or implied by a correct answer

(M1)

Answer = 0.83 to 0.84

AWFW

(A2)

3

(b) (i) $n = 5$ $p = 0.4$

Mean, $\mu = np = 2$

CAO

B1

Variance, $\sigma^2 = np(1 - p) = 1.2$

Use of $np(1 - p)$ even if SD

M1

Standard deviation = $\sqrt{1.2}$
or = 1.09 to 1.1

CAO
AWFW

A1

3

(ii) Mean $(\bar{x})$ = 2

CAO $\sum x$ *= 26*

B1

Standard Deviation (s_n, s_{n-1})
= 1.1 to 1.16

$\sum x^2$ *= 68*

AWFW (1.1094 or 1.1547)

B2

If neither correct but use of mean $(\bar{x}) = \frac{\sum x}{13}$

(M1)

3

(iii) Means are same and SDs are similar/same

Means are same but SDs are different

Must have scored full marks in (b) (i) and (b) (ii)

B1

so

↑Dep↑

Trina's claims appear valid / invalid

B1

2

[13]

Q18.

(a) (i) B(20, *p*)

Use of in (a)

M1

P(O < 10 | p = 0.40) = 0.872 to 0.873

AWFW 0.8725

A1

2

(ii) P(A = 8) = $\binom{20}{3}(p)^3(1-p)^{17}$

Any value of p with 0 < p < 1

M1

P(A = 8 | p = 0.20) =

$\binom{20}{3}(0.28)^3(0.72)^{17}$

Fully correct expression

A1

= 0.093 to 0.095
AWFW *0.0940*

A1
3

(iii) $P(4 < B < 8 \mid p = 0.20) =$

$P(B \leq 7 \text{ or } 8) = 0.9679 \text{ or } 0.9900$

M1

minus $P(B \leq 4 \text{ or } 3) = 0.6296 \text{ or } 0.4114$

M1

= 0.338 to 0.34
AWFW *0.3383*

A1

or

B(20, 0.2) expressions stated for at least 3 *of* B = 4, 5, 6, 7 & 8

(M1)

Answer
Or implied by a correct answer

(A2)
3

(b) Mean, $\mu = np = 500 \times 0.12 = 60$
CAO

B1

Variance, $\sigma^2 = np(1 - p) = 60 \times 0.88 = 52.8$
CAO

B1
2

[10]

Q19.

(a) (i) B(50, 0.2)
Use of in (a)

M1

$P(R \leq 15) = 0.969$ to 0.97

AWFW *0.9692*

A1

2

(ii) $P(R = 10) = P(R \leq 10) - P(R \leq 9)$

Stated or implied

or

M1

$P(R = 10) = \binom{50}{10}(0.2)^{10}(0.8)^{40}$

Stated or implied

$= 0.5836 - 0.4437 = 0.139$ to 0.141

AWFW *0.1399*

A1

2

(iii) $P(5 < R < 15) = F(R \leq 14$ or $15) = 0.9393$ or 0.9692

Accept values to 3 dp

M1

minus $P(R \leq 5$ or $4) = 0.0480$ or 0.0185

Accept values to 3 dp

M1

$= 0.89$ to 0.893

AWFW *0.8913*

A1

or

B(50, 0.2) expressions stated for at least 3 of $5 < R < 15$

Or implied by a correct answer

M1

Answer

(A2)

3

(b) Mean, $\mu = np = 50 \times 0.2 = 10$

Either; CAO

B1

or

Estimate of p, $\hat{p}$ = 0.21

Variance, $\sigma^2 = np\,(1 - p) = 10 \times 0.8 = 8$

CAO

B1

Mean or **Estimate of *p*** is similar to that expected

10.5 and 10 or 0.21 and 0.2 Either point

but

Variance (standard deviation) is **different** from that expected

20.41 and 8 or 4.5 and 2.8

B1

Reason to **doubt validity** of Sly's claim

Must be based on both 10 or 0.2 and 8 or on both 10 or 0.2 and 2.8 correctly

B1

4

[11]

Q20.

(a) B(15, 0.3)

use of in (a)

M1

(i) (Fewer than) half $\Rightarrow$ 7 or 7½ or 8

stated or implied

B1

Thus require P(K ≤ 7 or < 8)

used or implied by correct answer

M1

= 0.9495 to 0.9505

AWFW *(0.9500)*

A1

4

(ii) P(2 < K < 7) = 0.8689 or 0.9500

M1

minus 0.1268 or 0.2969

M1

= 0.7415 to 0.7425

AWFW (0.7421)

A1

or

B(15, 0.3) expressions stated for at least 3 terms within $2 \le K \le 7$

or implied by a correct answer

M1

Answer

A2

3

(b) (i) Mean, $\mu = np$ = 15 × 0.4 = 6

CAO

B1

Variance, $\sigma^2 = np(1-p)$ = 6 × 0.6 = 3.6

use of $\sigma^2 = np(1-p)$

M1

Standard deviation = $\sqrt{3.6}$ = 1.89 to 1.9

AWFW; or equivalent

A1

3

(ii) Mean, $\bar{x}$ = 6

CAO ($\Sigma x = 60$)

CSO if evidence of np(1–p) or 1.9

B1

Standard deviation, s or σ = 2.82 to 2.99

AWFW; or equivalent. ($\Sigma x^2 = 440$)

B1

2

(iii) Means are same/equal

ft on 2 means; accept $\frac{6}{15} = 0.4$
if not contradicted by $\bar{x}$ *in (ii)*

B1ft

Standard deviations are different
dependent on 2 correct SDs

B1 dep

Reason to doubt validity of Kirk's claim
dependent on 2 correct SDs

B1 dep

3

[15]

Q21.

(a) B(15, 0.3)
use of in (a)

M1

(i) P(K = 5) = P(K ≤ 5) – P(K ≤ 4)

P(K = 5) = $\binom{15}{5}(0.3)^5(0.7)^{10}$

may be implied

M1

= 0.7216 – 0.5155 = 0.2055 to 0.2065
AWFW *(0.2061)*

A1

3

(ii) (Fewer than) half $\Rightarrow$ 7 or 7½ or 8
stated or implied

B1

Thus require P(K ≤ 7 or < 8)
used or implied by correct answer

M1

= 0.9495 to 0.9505

AWFW *(0.9500)*

A1

3

(iii) P(2 < K < 7) = 0.8689 or 0.9500

M1

minus 0.1268 or 0.2969

M1

= 0.7415 to 0.7425

AWFW *(0.7421)*

A1

3

or

B(15, 0.3) expressions stated for at least 3 terms within $2 \le K \le 7$

or implied by a correct answer

M1

Answer

A2

(b) (i) Mean, $\mu = np$ = 15 × 0.4 = 6

CAO

B1

Variance, $\sigma^2 = np(1-p)$ = 6 × 0.6 = 3.6

use of $\sigma^2 = np(1-p)$

M1

Standard deviation = $\sqrt{3.6}$ = 1.89 to 1.9

AWFW; or equivalent

A1

3

(ii) Mean, $\bar{x}$ = 6

CAO *(Σx = 60)*

CSO if evidence of np(1–p) or 1.9

B1

Standard deviation, s or σ = 2.82 to 2.99

AWFW; or equivalent. (Σx^2 = 440)

B1

2

(iii) Means are same/equal

ft on 2 means; accept $\frac{6}{15} = 0.4$ *if not contradicted by* $\bar{x}$ *in (ii)*

B1ft

Standard deviations are different

dependent on 2 correct SDs

B1 dep

Reason to doubt validity of Kirk's claim

dependent on 2 correct SDs

B1 dep

3

[17]

Q22.

(a) (i) p = 0.4

Attempted use of B(7, 0.4) in (a)

M1

P($X \le 2$) = 0.419 to 0.421

AWFW (0.4199)

B1

(ii) P(X > 1 and X < 5) = P(2 ≤ X ≤ 4)

= P($X \le 4$)

Identification of ***at least*** *2, 3 and 4*

M1

– P($X \le 1$)

Identification of ***exactly*** *2, 3 and 4*

M1

= 0.9037 – 0.1586 = 0.744 to 0.746

AWFW (0.7451)

A1
5

(b) P(Y = 7) = $\binom{n}{7}(0.4)^7(0.6)^{n-7}$

Correct expression for B(7; n, 0.4) with n ≠ 7

M1

= $\binom{28}{7}(0.4)^7(0.6)^{21}$

Fully correct expression may be implied

A1

= 0.0425 to 0.0427

AWFW (0.042556)

A1
3

(c) Mean = np = 2.8

CAO

B1

SD = $\sqrt{np(1-p)}$ = $\sqrt{1.68}$

= 1.29 to 1.31

AWFW

B1
2

(d) (i) Mean = 2.8

CAO Σfx = 140

B1

SD = 2.24 to 2.27

AWFW $\Sigma fx^2 = 644$

B2

$s^2_{n-1} = 5.14$ to 5.15 and $s^2_{n-1} = 5.04$

Substitution of values into correct formula for variance or SD or
SD = 5.03 to 5.15 AWFW

M1

3

(ii) Means are the same

B1ft

SDs differ greatly

ft on (c) and (d)(i);
ft on (c) and (d)(i)
but must be s with σ or s^2 with σ^2

B1ft

Thus answers do not support Aaron's belief

Dependent on B1 above CAO

B1

3

[16]

Q23.

(a) Volume $X \sim N(56, 2.5^2)$

P (X < 60) = P $\left(Z < \frac{60-56}{2.5}\right)$

Standardising (59.5, 60 or 60.5) with 56

and ($\sqrt{2.5}$, 2.5 or 2.5^2) and/or (56 – x)

M1

(i) = P (Z < 1.6)

CAO; ignore sign

A1

= 0.945

AWRT (0.94520)

A1

3

(ii) P (50 < X < 60) = (i) – P (X < 50)

Or equivalent

M1

= (i) – P(Z < –2.4) = (i) – [1 – Φ(2.4)

Area change

m1

$= 0.94520 - (1 - 0.99180) = 0.937$

AWRT (0.93700)

A1 **3**

(iii) P (X = 55) = 0

CAO

B1 **1**

(b) 98% $\Rightarrow$ z = 2.05 to 2.06

AWFW; ignore sign (–2.0537)

B1

$z = \frac{100 - \mu}{3.4}$

Standardising 100 with μ & 3.4
Allow (μ –100)

M1

Thus $\frac{100 - \mu}{3.4} = -2.0537$

Equating z-term to z-value
Not using 0.98, 0.02 or |1 – z|

M1

Thus μ = 107

AWRT

A1 **4**

[11]

Q24.

(a) (i) Mean ($\bar{x}$) = 24.7 to 25.7

AWFW (25.2)

B2

Standard Deviation (s_n, s_{n-1})
= 16.7 to 17.7

AWFW (17.1474 or 17.2338)

B2

MPs (x): 5.5, 15.5, 23, 28, 33, 38, 45.5, 75.5

At least 4 correct

(B1)

Mean ($\bar{x}$) = $\frac{\sum fx}{100}$

Use of

(M1) 4

(b) Data is skewed or not symmetric
Discrete data or counts

One valid reason

B1 1

(Mean – 2 × SD) < 0 $\Rightarrow$ negative counts

(c) (i) Since sample size large (n > 30)
can use Central Limit Theorem

Either point

B1 1

(ii) Mean = μ

CAO; not $\bar{x}$ *or its value*

B1

Variance = $\frac{\sigma^2}{100}$

Accept $\frac{\sigma^2}{n} or \frac{(their SD)^2}{100}, etc$

B1 2

(d) 99% $\Rightarrow$ z = 2.57 to 2.58

AWFW (2.5758)

B1

CI for μ is $\bar{x} \pm z \times \frac{(\sigma \text{ or } s)}{\sqrt{n}}$

Use of

Must have ($\div \sqrt{n}$) *with* $n > 1$

M1

Thus $25.2 \pm 2.5758 \times \frac{17.1 \text{ or } 17.2}{\sqrt{100}}$

ft on $\bar{x}$ *, z and s > 0; not on n*

A1ft

(20.8, 29.6)

AWRT

A1

4

(e) UCL < 30

ft on CI

B1ft

so

↑ dep

Reject claim that $\mu > 30$ B1ft

ft on CI

7/100 or 7% of $X > 50$ (from table)

CAO

B1

so

↑ dep

Reject claim that often $X > 50$

CAO

B1

4

[16]

Q25.

Weight, $X \sim N(205, 25^2)$

(a) (i) $P(X < 250) = P\left(Z < \dfrac{250 - 205}{25}\right)$

Standardising (249.5, 250 or 250.5)

with 205 and ($\sqrt{25}$, 25 or 25^2) and/or $(205 - x)$

M1

$= P(Z < 1.8)$

CAO; ignore sign

A1

$= 0.964$

AWRT (0.96407)

A1

3

(ii) $P(200 < X < 250) = \text{(i)} - P(X < 200)$

Or equivalent

M1

$= \text{(i)} - P(Z < -0.2) = \text{(i)} - [1 - \Phi(0.2)]$

Area change

$= 0.96407 - (1 - 0.57926) = 0.543$

AWRT (0.54333)

A1

3

(b) (i) $(100 - 30)\% = 70\% \Rightarrow z = 0.524$ to 0.525

AWFW; ignore sign (–0.5244)

B1

Thus $\dfrac{s - 205}{25} = -0.5244$

Equating z-term, involving 205 and 25,
to z value
Not using 0.3, 0.7 or |1 – z|
Allow (205 – s)

M1

Thus s = 191.9

AWRT

A1

(ii) (100 – 20)% = 80% $\Rightarrow$ z = 0.841 to 0.842

AWFW, ignore sign (0.8416)

B1

Thus $\frac{m - 205}{25}$ = 0.8416

Only if not awarded in (i)
Not using 0.2 or 0.8
Allow (205 – m)

(M1)

m = 226.0

AWRT; accept 226

A1

5

(c) 90% $\Rightarrow$ z = 1.28

AWRT; ignore sign (1.2816)

B1

z = $\frac{200 - 175}{\sigma}$

Standardising 200 with 175 & σ
Do not allow 175 – 200

M1

Thus $\frac{25}{\sigma}$ = 1.2816

Equating z-term to z-value

Not using 0.9 or 0.1

m1

Thus σ = 19.5

AWRT

A1

4

[15]

Q26.

(a) (i) <u>B(n, 0.07)</u>

Use of in (a)

M1

P(X = 2) = $\binom{17}{2}(0.07)^2(0.93)^{15}$

Fully correct expression
May be implied

A1

= 136 × 0.0049 × 0.33670
= 0.224 to 0.225

AWFW (0.22438)

A1

3

(ii) P ($X \leq 5$) | B(50, 0.07))

Attempted; tables or
formula (≥ 3 terms stated)
May be implied

M1

= 0.865

AWRT (0.8650)

A1

2

(b) <u>B (50, 0.55)</u>

P ($Y \geq 30$) = P ($Y' \leq$ 20)

Change from Y to Y′
Must be clear evidence

M1

with p = 0.45

Stated or implied

A1

= 0.286

AWRT (0.2862)

A1

3

(c) (i) Estimate of $p = \frac{10}{50} = 0.2$

CAO

B1

1

(ii) Estimate of SD (X) = $\sqrt{np(1-p)}$

Use of; accept no √

M1

$= \sqrt{50 \times 0.2 \times 0.8} = \sqrt{8}$

= 2.82 to 2.83

AWFW; accept $\sqrt{8}$

A1

2

(iii) SD(X) less than 6.8

Comparison

or

ft on (c)(ii)

M1ft

V (X) less than 46.24

Must be like with like

Not a reasonable assumption

ft on (c)(ii) and like with like comparison

A1ft

2

[13]

Q27.

(a) Length $X \sim N(48, 8^2)$

$P(X = 60) = 0$

CAO

B1

1

(b) $P(X = 60) = P\left(Z < \frac{60 - 48}{8}\right)$

standardising (59.5, 60 or 60.5) with 48 and $\left(\sqrt{8}, 8 \text{ or } 8^2\right)$ *and/or* $(48 - x)$

M1

$P = (Z < 1.5) = 0.933$

AWRT (0.93319)

A1

2

(c) 5% of 48 = 2.4

CAO; OE

B1

$P(48 - x < X < 48 + x)$

attempt at the probability of an interval, symmetric about 48

M1

$= P(-0.3 < Z < 0.3)$

CAO 0.3

A1

$= \Phi(0.3) - (1 - \Phi(0.3))$

area change

m1

$= 2 \times 0.61791 - 1$

$= 0.235$ to 0.236

AWFW (0.23582)

A1

5

[8]

Q28.

(a) $Y \sim$ Geo (0.8)

B1

1

(b) P(Y = 5) = $(0.2)^4(0.8)$

M1

= 0.00128 =

A1

2

(c) E(Y) = $\frac{1}{p} = \frac{1}{0.8} = 1.25$

CAO

B1

Var(Y) = $\frac{q}{p^2} = \frac{0.2}{0.64} = \frac{5}{16}$

= 0.3125

CAO

B1

2

[5]

Q29.

(a) n = 18 p = 0.15

P(Car = 2) =

binomial used in (a) or (b)

M1

$\binom{18}{2}(0.15)^2(0.85)^{16}$

correct expression

A1

= 0.255 to 0.256

AWFW *(0.2556)*

A1

3

(b) $n = 50$ $p = 0.15$
P $(5 < Car < 10)$ =

P $(Car \le 9)$
–P $(Car \le 5)$
Use of ≤ 9 or (6, 7, 8, 9)

M1

Use of – & ≤ 5 or

M1

(4 correct terms added)

= 0.7911 – 0.2194 = 0.571 to 0.572
AWFW *(0.5717)*

A1

3

(c) $n = 900$ $p = 0.15$
$\mu = 900 \times 0.15 = 135$
CAO

B1

$\sigma^2 = 900 \times 0.15 \times 0.85 = 114$ to 115
114.75 (σ = 10.65 to 10.75 AWFW)

B1

P$(Car \le 150)$ = P$(Car < 150.5)$
+ 0.5

B1

$$= \text{P}\left(Z < \frac{150.5 - 135}{\sqrt{114.75}}\right)$$

standardising (149.5, 150, 150.5) using their μ

& their $\sqrt{\sigma^2}$ or correct values

M1

= P(Z<1.45) = Φ(l.45) = 0.926 to 0.927
AWFW *(0.92647)*

A1

5

(d) p not 0.15 (value for cars, not all vehicles)
Vehicles not independent

E1

1

[12]

Q30.

$n = 20$ $p_0 = 0.4$ $p_R = 0.6$

(a) P(O = 10) or P(R = 10)

identification of B (20, 0.4 OR 0.6) in (a) or (b)

B1

$= \binom{20}{10}(0.4)^{10}(0.6)^{10}$

use of formula with $^{20}C_r$, $p + q = 1$, Σpowers = 20 or tables ($\leq 10 - \leq 9$)
(M0 for normal approx)
(0.117 × 2 $\Rightarrow$ B1, M1, AO)

M1

184756×0.000104857 × 0.006046617 = 0.8725 – 0.7553
= 0.117

AWRT

A1

3

(b) $P(O < 10) = P(0 \leq 9)$

attempt at ≤ 9 (or ≥ 11)
(M0 for normal approx)

M1

= 0.755

AWRT

A1

2

(c) Sample size or n is not fixed

OE
n = 0 not available $\Rightarrow$ E1
Geometric $\Rightarrow$ E0

E1

1

[6]

Q31.

(a) (i) Binomial $n = 50$ $p = 0.4$

Binomial

B1

P(20 or fewer) = 0.5610

$n = 50$ $p = 0.4$

B1

0.561 (0.5605 ~ 0.5615)

B1

3

(ii) 20 or fewer invigorating $\rightarrow$ 30 or more relaxing

reasonable attempt to express in terms of number of relaxing cubes or method for calculating 20 or fewer invigorating

M1

P(30 or more) = 1 – P(29 or fewer)

P(30 or more) = 1 – P(29 or fewer)

m1

1 – 0.9966
= 0.0034

0.0034 (0.00335 ~ 0.00345)

A1

3

(iii) More relaxing $\rightarrow$ 26 or more relaxing

M1

P(26 or more) = 1 – P(25 or fewer)
= 1 – 0.9427

reasonable attempt to express more relaxing in terms of number of relaxing cubes

= 0.0573

0.0573 (0.057 ~ 0.0574)

A1

2

(b) (i) Not binomial, n not fixed

not binomial

M1

n not fixed

A1

(ii) Not binomial, p not constant/not

not binomial

M1

independent

reason

A1

4

[12]

Q32.

(a) $z = \frac{20 - 26}{8} = -0.75$

method for z – ignore sign

M1

probability no need to refill

completely correct method

M1

1 – 0.77337 = 0.227

0.227 (0.226 to 0.227)

A1

$z = \frac{40 - 26}{8} = 1.75$

probability exactly one refill i.e. between

M1

20 and 40 = 0.95994 – 0.2263 = 0.733

correct method

0.733 (0.733 to 0.734)

A1

5

(b) $0.22623^5 = 0.000598$

M1

0.000598 (0.00059 to 0.00061)

A1

2

(c) $\mu - 0.9945\sigma = 20$

0.9945 (0.994 to 0.995)

B1

$\mu - 1.175\sigma = 40$

1.175 (1.17 to 1.18)

B1

Good attempt at equations - ignore sign

M1

Completely correct equations

m1

2.169σ

Method of solution

m1

$\sigma = 9.222$

A1

6

$\mu = 29.2$

σ = 9.222 (9.21 to 9.23)
and μ= 29.2 (29.1 to 29.3)

[13]

Q33.

(a) (i) Binomial $n = 8$ $p = 0.3$

Binomial
8, 0.3

B1 B1

P(2 or fewer) = 0.552

0.552 (0.551, 0.5525)

B1

(ii) P(2) = 0.5518 – 0.2553 = 0.2965

P(2 or fewer) – P(1 or fewer) or use of correct formula

M1

0.2965(0.296, 0.297)

A1

(iii) P(>3) = 1 – 0.8059 = 0.194

P(>3) = 1 – P(3 or fewer) or use of correct formula
0.194(0.193, 0.195)

A1

7

***sc** B1 0.448 (0.448 , 0.449)*

(b) No, *n* not constant/probabilities not random/not independent/ 0,1 not possible outcomes

No

Reason

M1 A1

2

(c) No, *p* not constant/ not independent

No

Reason

M1 A1

2

[11]

Q34.

(a) (i) Binomial $n = 5$ $p = 0.15$

Binomial

B1

$n = 5$ $p = 0.15$

B1

p (2 or fewer) = 0.973

0.973 (0.973 to 0.974)

B1

3

(ii) *p* (more than 3) = 1 – 0.9978

p(>3) = 1 – P (3 or fewer)

M1

= 0.0022

0.0022(0.00215 to 0.002250) Allow 0.002

A1 2

(iii) mean = 5 × 0.15 = 0.75

*0.75 **cao***

B1

s.d = $\sqrt{5} \times 0.15 \times 0.85 = \sqrt{0.6375}$

Method – allow variance if called variance even if incorrect late variance = 0.6375

M1

= 0.798

0.798 (0.798 to 0.8)

***sc** Use of n = 7 in this question*

(a)(i) Binomial n = 7 p = 0.15

0.9262 3

(ii) 0.0121 2 MR – 1

(iii) 1.05 0.945 3

Total 7

No further allowance in (b) and (c) (if they continue with n = 7 this would be a second misread and only (b)(ii) is affected).

A1 3

(b) (i) mean = 0.75

*0.75 **cao***

B1

s.d = 1.24 or 1.23

1.24 (1.23 to 1.2) allow M1A1 if method shown

B2 3

(ii) Proportion = $\frac{0.75}{5}$ = 0.15

Their mean divided by 5

E1f.t. 1

(c) (i) Not plausible
s.d. much larger than for binomial

B1

s.d. too large – allow arguments based on probabilities *calculated in (a)(i) and (ii)*

E1f.t.

2

(ii) p not constant – some pupils more likely to be late than others

p not constant / not independent

E1

late arrivals not independent – may be due to weather/transport difficulties etc

For 2 marks, need 2 reasons with at least one in context.

E1

2

[16]

Q35.

(a) $z_1 = \dfrac{125 - 129}{2.5} = -1.6$

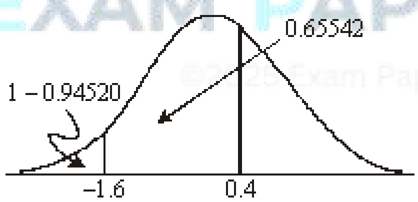
Method for z ignore sign; ignore attempt at continuing correction

M1

$z_2 = \dfrac{130 - 129}{2.5} = -1$

both z's correct sign

m1



Probability between 125 and 130 = 0.65542 – (1 – 0.94520) = 0.601

A1

5

(b) (i) 127.5

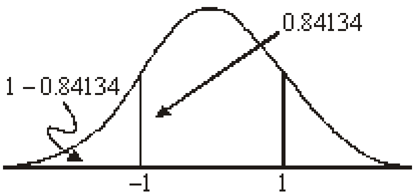
127.5 or 127 or 128 only

B1

1

(ii) $z_1 = \dfrac{130 - 127.5}{2.5} = 1$

$z_2 = \dfrac{125 - 127.5}{2.5} = -1$



Their mean

M1

Probability between 125 and 130

= 0.84134 –(1 – 0.84134) = 0.683

Completely correct method (dependent on M1 and 127.5)

0.683 (0.68 – 0.685)

m1 A1

3

(c) 125 + 2.3263 × 2.5 = 130.8

2.3263 (2.32 – 2.33)

their z × 2.5

m1 correct method

130.8 (130.8 – 130.82) or 131

B1 M1 m1 A1

4

[12]